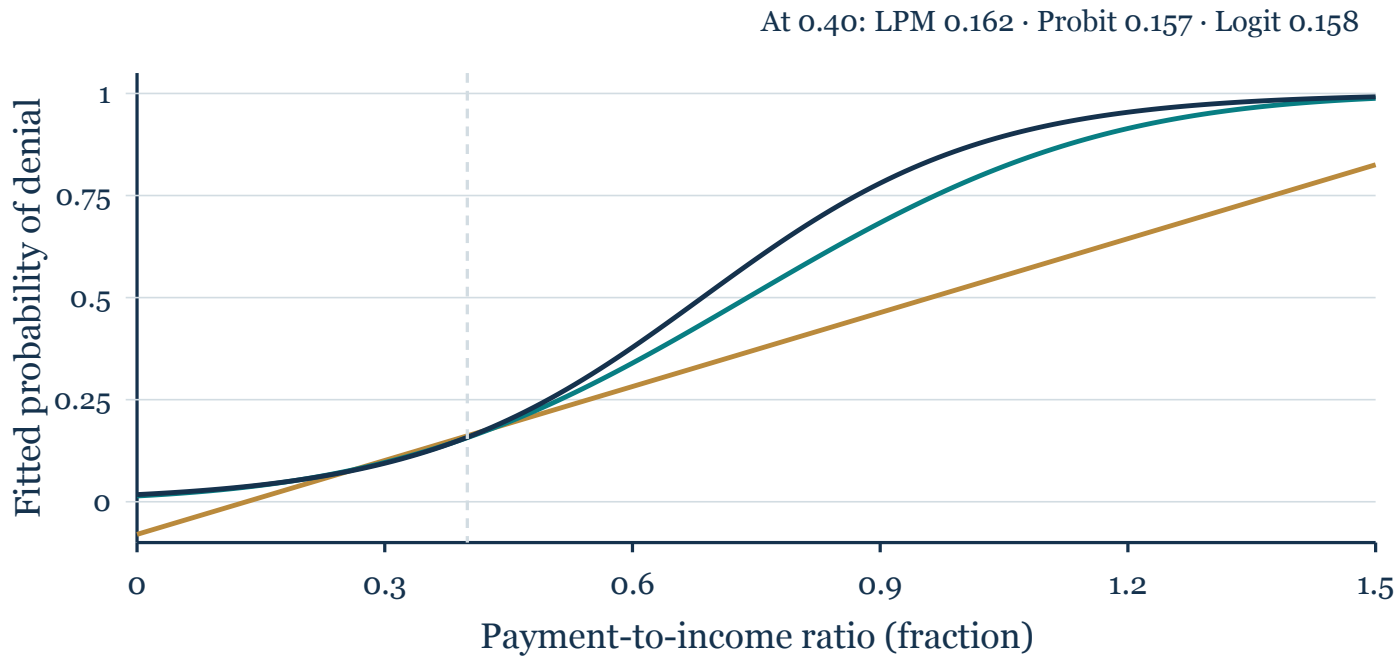
# How does payment burden predict mortgage denial?

A zero-or-one outcome is an event indicator.




Actual fitted LPM, probit and logit predictions from the same 2,380 applications. Payment-to-income is a fraction; these are descriptive fits.

> Linear probability model: straight-line probability fit; logit and probit: bounded probability fits

# The conditional mean of an indicator is a probability

This gives the linear model a useful starting interpretation.

> $E(Y \mid X) = P(Y = 1 \mid X)$ for $Y \in \{0, 1\}$

> LPM: $P(Y = 1 \mid X) \approx a + bX$

# A probability change is not a relative percent change

Keep percentage points and percent distinct.

> $0.23 - 0.18 = 0.05 = 5$ percentage points

> Relative percent: $100(0.05/0.18) \approx 27.8\%$

# Read the actual LPM coefficient

Use a feasible ratio change, not an entire unit by default.

```
. regress deny pir, vce(robust)

------------------------------------------------------------------------------
             |               Robust
        deny | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   .6035349   .0984826     6.13   0.000     .4104144    .7966555
       _cons |  -.0799096   .0319666    -2.50   0.012    -.1425949   -.0172243
------------------------------------------------------------------------------
```

# The LPM error variance is generally unequal

Robust SEs address uncertainty, not invalid fitted probabilities.

> $$\mathrm{Var}(Y \mid X) = p(X)\{1 - p(X)\}$$

# Probability scale

Apply the idea before reading the solution.

**Practice**

An LPM predicts 0.18 at one covariate vector and 0.23 at another. State the change in percentage points and relative percent.

# Probability scale: explanation

Connect the calculation to the empirical claim.

> The change is 0.05, or 5 percentage points. Relative to 0.18 it is $100 \times 0.05/0.18 = 27.78\%$. These are different scales. The contrast is predictive unless identification assumptions support a causal claim.

# Logit and probit keep fitted probabilities in range

A link function maps an unrestricted index into zero to one.

> Logit: $p(X) = \Lambda(a + bX) = \dfrac{1}{1 + e^{-(a+bX)}}$

> Probit: $p(X) = \Phi(a + bX)$

> $\Lambda$: logistic CDF; $\Phi$: standard-normal CDF

# A logit coefficient changes log odds

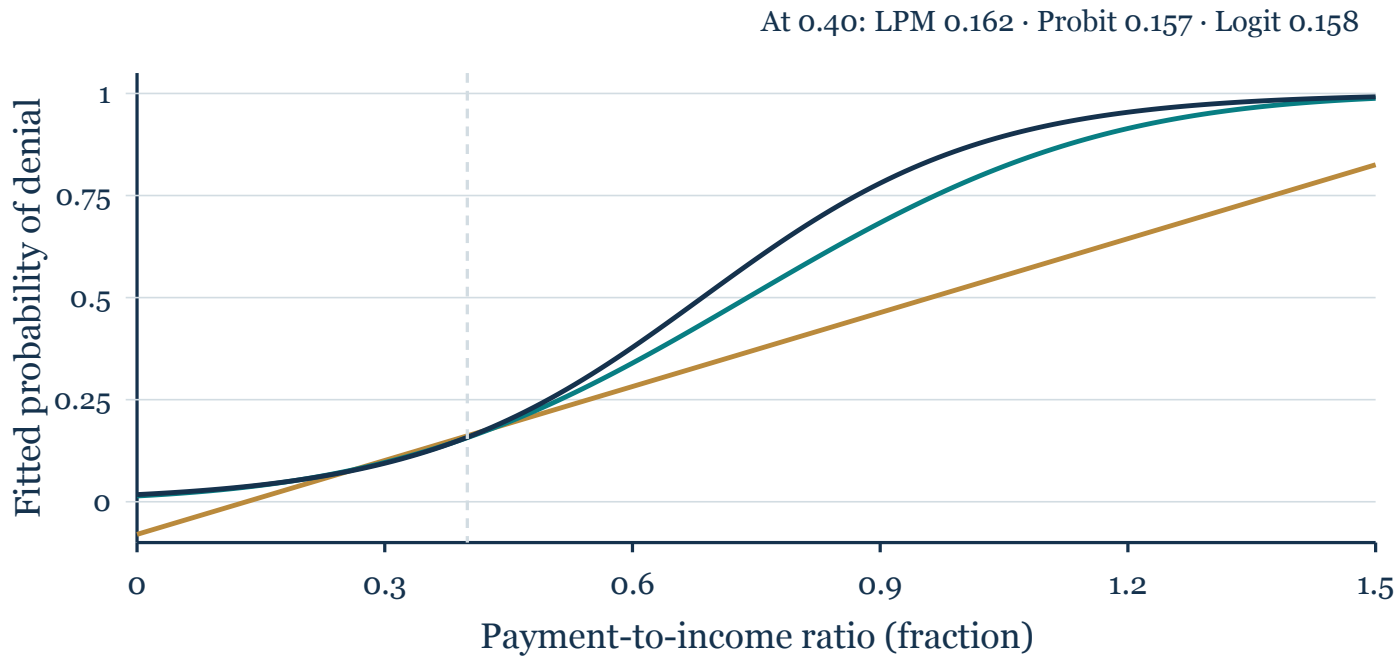
Translate it before discussing probabilities.

> $\log\{p/(1-p)\} = a + bX$

> Odds ratio for $\Delta X$: $\exp(b\Delta X)$

# Probit also uses an index coefficient

Its probability effect depends on the reference density.




Actual fitted LPM, probit and logit predictions from the same 2,380 applications. Payment-to-income is a fraction; these are descriptive fits.

$\frac{\partial p}{\partial X} = b\phi(a + bX)$ for probit

$\phi$: normal density; $\partial p/\partial X$: local probability slope

# Maximum likelihood chooses probabilities that support the observed events

Define the objective before interpreting the estimator.

> $L(\theta) = \prod_i p_i(\theta)^{Y_i}\{1 - p_i(\theta)\}^{1-Y_i}$

> $\ell(\theta) = \sum_i [Y_i \log p_i + (1 - Y_i)\log(1 - p_i)]$

> $\theta$: coefficients; $L$: likelihood; $\ell$: log likelihood

# Compare the actual fitted probit and logit scales

Their coefficients are not directly comparable magnitudes.

```
. probit deny pir, vce(robust)

------------------------------------------------------------------------------
             |               Robust
        deny | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   2.967907   .4653217     6.38   0.000     2.055893    3.879921
       _cons |  -2.194159   .1649761   -13.30   0.000    -2.517506   -1.870812
------------------------------------------------------------------------------
```

# Predict probabilities at meaningful ratio values

State which covariates are changed or held fixed.

```
. margins, at(pir=(.3 .4 .5))

Adjusted predictions                                  Number of obs = 2,380
Model VCE: Robust

Expression: Pr(deny), predict()
1._at: pir = .3
2._at: pir = .4
3._at: pir = .5

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |   .0961532   .0069017    13.93   0.000      .082626    .1096803
          2  |   .1569683   .0101166    15.52   0.000     .1371402    .1767965
          3  |   .2387884    .024485     9.75   0.000     .1907987    .2867781
------------------------------------------------------------------------------
```

# Logit coefficient

Apply the idea before reading the solution.

**Practice**

Why is a logit coefficient of 0.5 not a 50-percentage-point increase in probability?

# Logit coefficient: explanation

Connect the calculation to the empirical claim.

> It is a change in the log odds per unit of the regressor, conditional on the specification. The probability change depends on the starting index and other covariates. Report an explicit prediction contrast or marginal effect to discuss probabilities.

# A marginal effect is a local probability slope

For an indicator, use a discrete comparison instead.

> Logit: $\partial p / \partial X = b\, p(1-p)$

> Binary $D$: $p(D=1, X) - p(D=0, X)$

# An average marginal effect averages individual slopes

The effect at the average covariates is another calculation.

```
. margins, dydx(pir)

Average marginal effects                                  Number of obs = 2,380
Model VCE: Robust

Expression: Pr(deny), predict()
dy/dx wrt:  pir

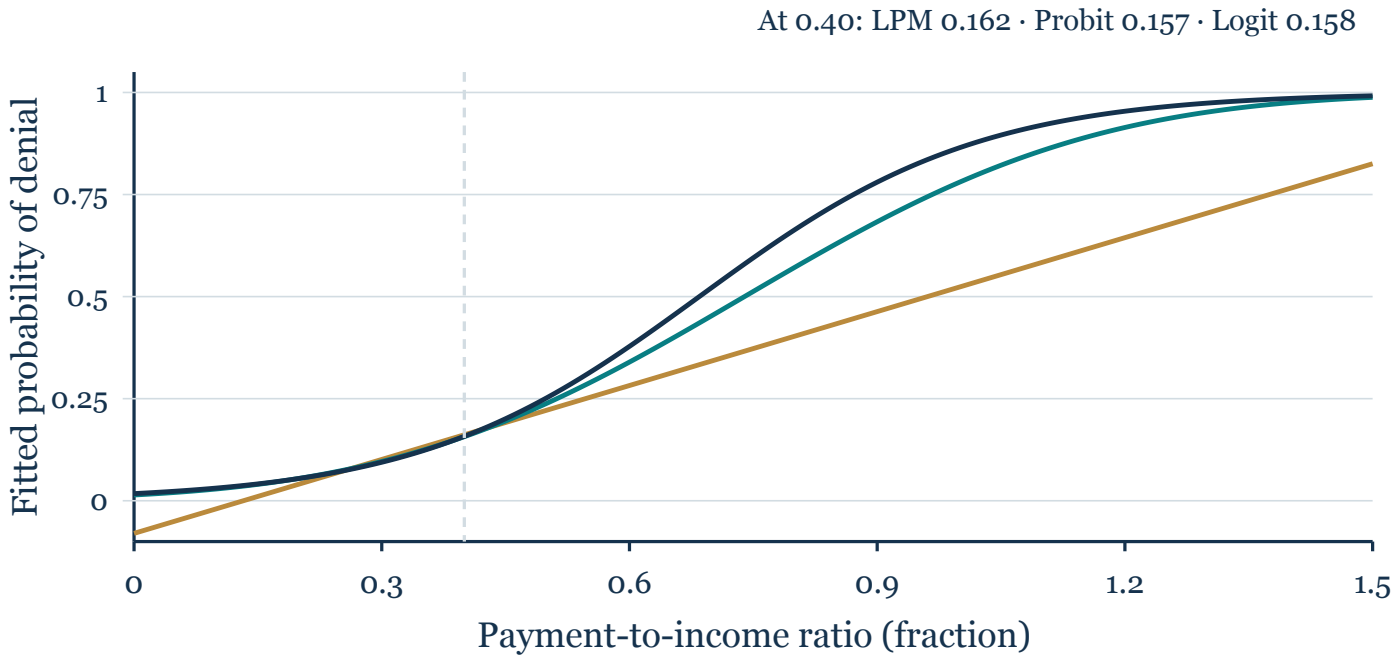
------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   .5664828   .0866687     6.54   0.000     .3966153    .7363504
------------------------------------------------------------------------------
```

$$\text{AME} = \frac{1}{n} \sum_i \left. \frac{\partial p(X)}{\partial X_j} \right|_{X = X_i}$$

$X_j$: selected regressor; $X_i$: applicant $i$'s covariates

# A finite probability change can differ from a local approximation

Evaluate both endpoints when the proposed change is substantial.




Actual fitted LPM, probit and logit predictions from the same 2,380 applications. Payment-to-income is a fraction; these are descriptive fits.

# Prediction performance needs more than a significance test

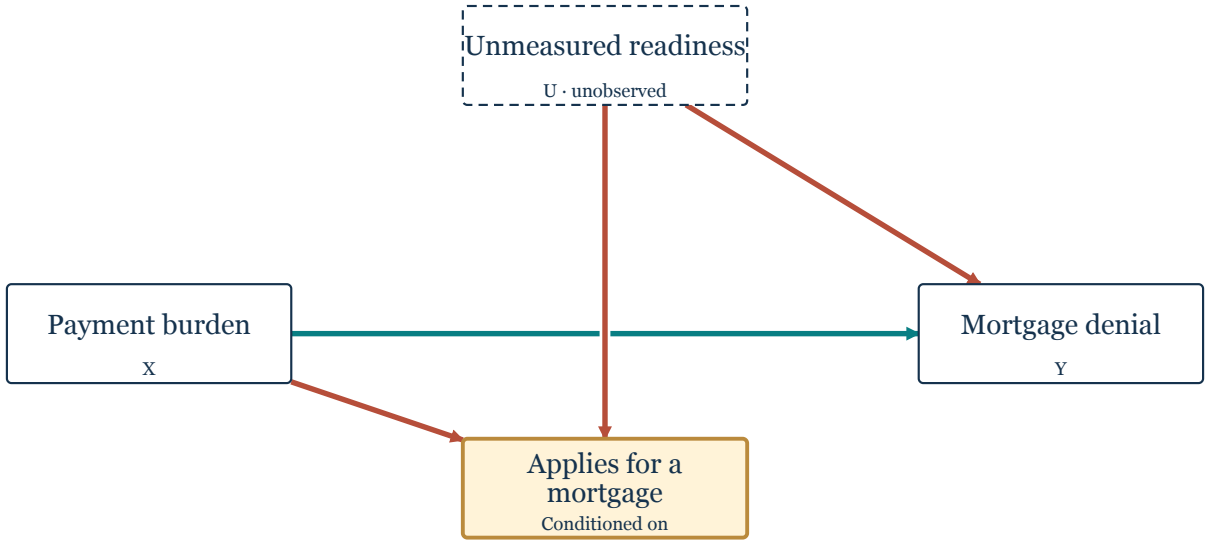
Check the probability model against its task.

> A coefficient test asks about a parameter under the model; it does not fully assess prediction quality.

> For a rare event, predicting zero for everyone can produce high classification accuracy while providing little useful probability information.

# A fitted denial gap is not automatically a treatment effect

Selection and omitted applicant characteristics can remain.




1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

# Write a compact applied research memo

Connect the question, data, model, assumptions and result.

> A useful memo identifies the decision or economic question, defines the observation and variables, explains the comparison and reports a meaningful magnitude with uncertainty.

> For the mortgage example, use a payment-burden contrast in percentage points and explain how it was obtained from the fitted model.

# Average marginal effect

Apply the idea before reading the solution.

**Practice**

Distinguish an average marginal effect from the marginal effect evaluated at the mean covariate vector.

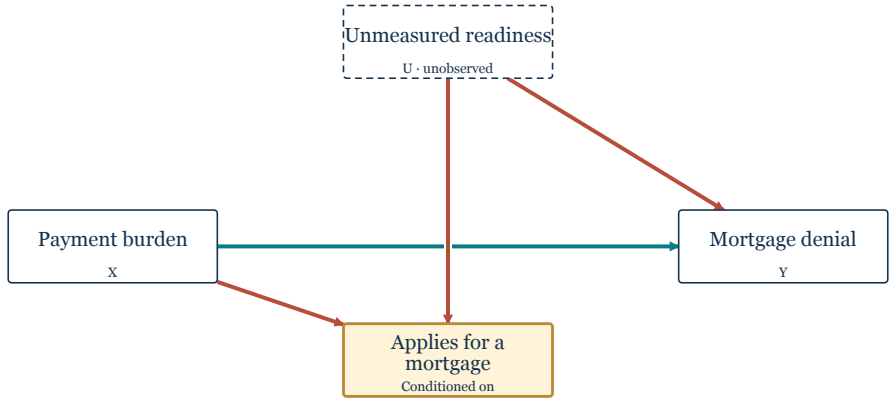
# Average marginal effect: explanation

Connect the calculation to the empirical claim.

> The AME computes an effect at each observation's covariates and averages those effects. The effect at means computes one effect at the vector of average covariates. In nonlinear models the two generally differ, and the average vector can describe no actual person.

# DAG practice: selected applicants and prediction

Trace the routes before choosing a control.




1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

## Practice

All observed records are mortgage applicants. In the diagram, payment burden and unmeasured readiness affect applying, while readiness also affects denial. Trace the route opened by selecting applicants. Does this make within-applicant prediction meaningless, or does it change the causal claim that needs defending?

# DAG practice: selected applicants and prediction: explanation

Explain what conditioning changes in this stated graph.

> Selecting on application opens $X \rightarrow S \leftarrow U \rightarrow Y$, because application is a common consequence of burden and readiness. A causal comparison of burden can then mix its relationship with the outcome and selection through readiness. Predicting denial conditional on burden within the observed applicant population remains a legitimate descriptive task. A causal change in burden, or a conclusion about all potential borrowers, requires a clear target and additional identification and selection assumptions. A bounded logit or probit function does not itself close the path.

# Compare probabilities on one common scale

Reproduce the evidence, then explain its economic meaning.

**Practice**

Run meeting_15.do using mortgages.csv. Read the fitted probit denial probabilities at payment-to-income ratios 0.30 and 0.40, and calculate their difference in percentage points. Explain why multiplying the probit coefficient by 0.10 does not generally give this change. State whether the result measures a causal lending response.

# Compare probabilities on one common scale: explanation

A reproducible result needs both a calculation and an interpretation.

> The fitted probit probabilities are 0.096153 at 0.30 and 0.156968 at 0.40. Their difference is 6.0815 percentage points. Both inputs are fractions, so the proposed change is ten percentage points of payment burden. The probit coefficient multiplies the linear index. The probability change also depends on the nonlinear cumulative-normal function and the starting index, so a coefficient-times-change shortcut does not give the finite probability contrast. These predictions use the stated one-regressor historical sample model. Without a credible account of borrower selection and other omitted lending determinants, report a descriptive prediction contrast rather than a causal response to changing an applicant's burden.

# Choose the method that addresses the empirical problem

A technical tool is useful when its role is explicit.

> Multiple regression accounts for stated observed differences.

> These roles can be combined, but none is a universal stamp of credibility.